AlMidrar_SiteSurvey

Introduction

The purpose of this networking site survey was to assess the current network infrastructure and performance of the Al-Midrar to identify potential areas for improvement. The survey was conducted by the Faizan, Shah Meer, and Kashif on Friday 21 July 2023.

Objectives

The primary objectives of the networking site survey were as follows:

- Evaluate the existing network topology, equipment, services, and configurations.
- Identify network performance issues and bottlenecks.
- Recommend improvements to enhance the overall network efficiency, cost, and reliability.

Methodology

- [Done] Initial Consultation: Discussions with the IT team to understand the current network setup, challenges faced, and objectives.
- [Done] Physical Inspection: On-site examination of the networking equipment, cabling, and network rooms.
- [Planed for next visit] Network Performance Testing: Utilizing specialized software to analyze network speed, latency, and throughput.
- [Planed for next visit] User Interviews: Obtaining feedback from employees regarding their network experience and pain points.

Current Setup and Network Infrastructure

- Buildings : 3 | Floors : 5 | Rooms : 13
- Network Topology: Mesh Wifi
- Network Equipment:
  - Wifi Routers: 13 (4*30mbps+9*30mbps = 430mbps)
- Network Configuration: Default
- List of clients : (Total 210+1)
  - Smartphones(Wifi) : 92
  - Laptops/PC(Wifi) : 92
  - CCTV Cameras(Wifi) : 26
  - Attendance Machine(unknown interface) : 1

Findings

1. All clients are connected via Wifi
2. 13 Internet connections
3. No Bandwidth policy is configured
4. Data is stored on local PCs
5. Mostly used services are: WhatsApp, zoom, video uploading

Building #	Floor#	Room #	Routers	Router WAN	Router Eth out Tx1000	ISP Bandwidth (mbps)	Mostly Used Services	Wifi Smartphones	Wifi Laptop/PC	Wifi CCTV CAMs	Ethernet Nodes	Total Nodes	Network Policies
1	1	1	Huawei Wifi EG8143A5*	Optical	4	40	WhatsApp + CCTV	8	8	2	0	18	Default
1	1	2	Huawei Wifi EG8143A5*	Optical	4	40	WhatsApp + CCTV	8	8	2	0	18	Default
1	1	3	Huawei Wifi EG8143A5*	Optical	4	30	WhatsApp + CCTV	8	8	2	0	18	Default
1	1	4	Huawei Wifi EG8143A5*	Optical	4	30	WhatsApp + Video Streaming Up + CCTV	8	8	2	0	18	Default
1	2	1	Huawei Wifi EG8143A5*	Optical	4	30	Video Streaming Up/Down + Zoom + CCTV	4	4	2	0	10	Default

Building #	Floor#	Room #	Routers	Router WAN	Router Eth out Tx1000	ISP Bandwidth (mbps)	Mostly Used Services	Wifi Smartphones	Wifi Laptop/PC	Wifi CCTV CAMs	Ethernet Nodes	Total Nodes	Network Policies
1	2	1	Huawei Wifi EG8143A5*	Optical	4	30	Video Streaming Up/Down + Zoom + CCTV	4	4	2	0	10	Default
1	2	2	Huawei Wifi EG8143A5*	Optical	4	30	Video Streaming Up/Down + Zoom + CCTV	4	4	2	0	10	Default
1	2	3	Huawei Wifi EG8143A5*	Optical	4	40	Video Streaming Up/Down + Zoom + CCTV	4	4	2	0	10	Default
1	3	1	Huawei Wifi EG8143A5*	Optical	4	40	WhatsApp + CCTV	14	14	2	0	30	Default
1	3	2	Huawei Wifi EG8143A5*	Optical	4	30	WhatsApp + CCTV	14	14	2	0	30	Default
1	3	3	Huawei Wifi EG8143A5*	Optical	4	30	WhatsApp + CCTV	14	14	2	0	30	Default
2	UNK	1	Huawei Wifi EG8143A5*	Optical	4	30	WhatsApp + CCTV	2	2	2	0	6	Default
3	UNK	1	Huawei Wifi EG8143A5*	Optical	4	30	WhatsApp + CCTV	0	0	2	0	2	Default
3	4	13	13		52	430		92	92	26	0	210	

Recommendations

- Shift all Laptop/Pc and CCTV from wifi to wired network
- Implement Quality of Service (QoS) policies to prioritize critical applications during periods of high traffic(required managed switches).
- Reduce the number of internet connections from 13 to 4-5 and upgrade bandwidth plans

Benefits of Wired Network Conversion

- Enhanced Network Performance: Wired connections generally provide faster and more reliable data transfer compared to wireless.
- Reduced Interference: Eliminating Wi-Fi signals reduces potential interference issues in crowded or congested environments.
- Enhanced Security: Wired networks offer higher security than wireless networks, as they are not susceptible to unauthorized access via Wi-Fi hacking.

Challenges and Considerations

- Disruptions during Installation: The conversion process may cause temporary disruptions to network services while cabling and equipment are installed.
- Cost: Converting to a wired network may involve significant upfront costs for cabling, switches, and other hardware.
- Physical Constraints: The office's physical layout may present challenges in running cables to certain areas.

Proposed Solutions

01 Convert with an existing Internet connection

- 12 Unmanaged network switches shall be added with each router
- All Laptop/PC and CCTV cams shall be converted into a wired network

Pros

- Cost-effective (infrastructure-wise)
- No new connections are required

Cons

- Hard to manage because using 13 different connections
- High running cost (13 connections)

Diagram

Required Hardware

- 12 16-ports unmanaged switches
- Cat 6 cable (need site survey)
- Channels for cable
- Power cable

02 Restructured the whole Infrastructure

- 4-5 internet connections required
- 4 Managed switches shall be installed on each floor of the building
- 10-12 Wifi Routers shall be installed in each room for smartphones(92+)

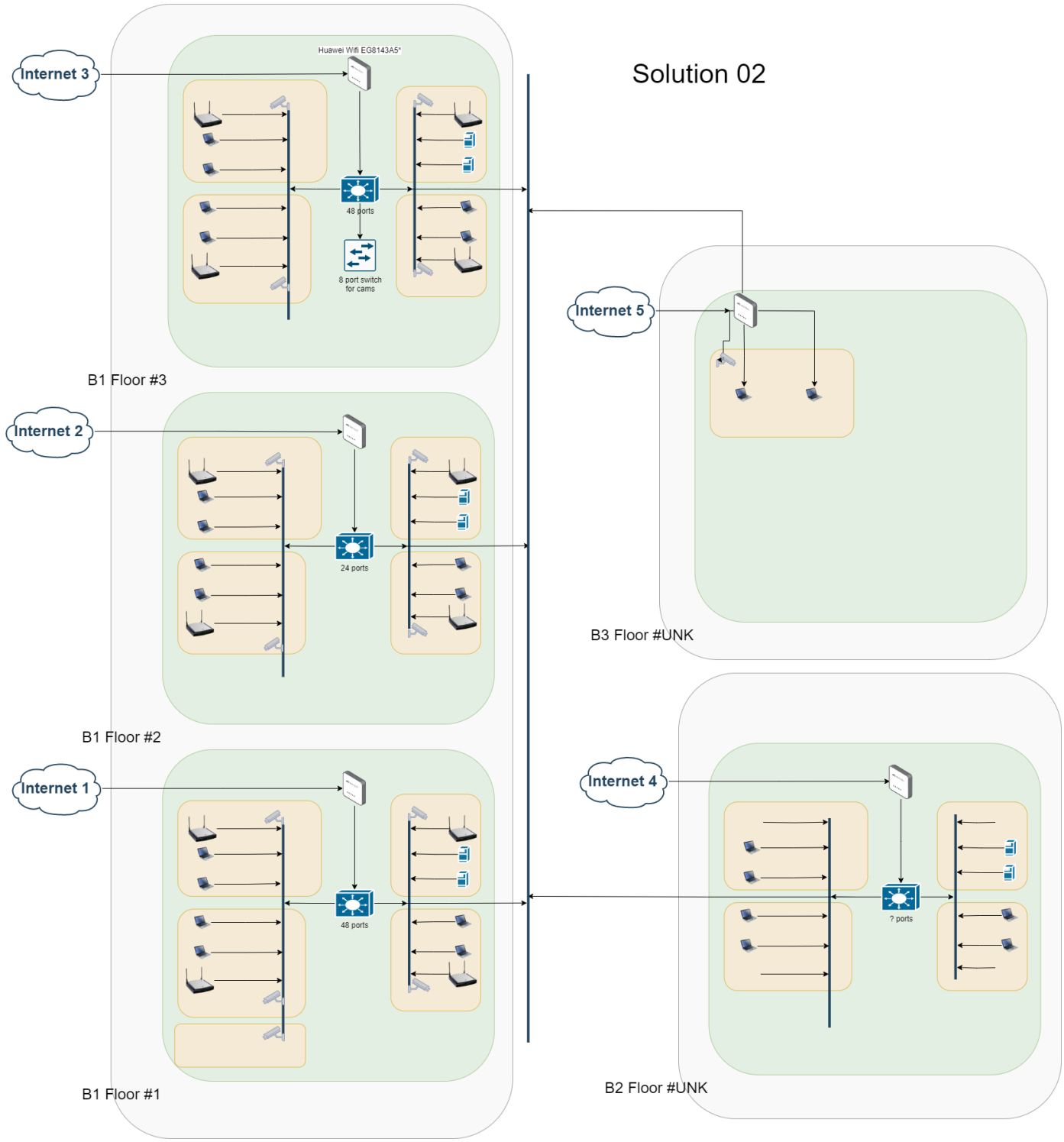
Pros

- 4-5 internet connections shall require instead of 13
- Low running cost
- Can manage resources of all nodes via console (bandwidth control, access control ....)
- Users can also access CCTV cams/nodes via the local network instead of using the internet across all floors*.

Cons

- High one-time Infrastructure cost
- Need dedicated network admin

Diagram



Required Hardware

- 2 48-port L3 switches
- 1 24-ports unmanaged switch
- 12 Tp-Link 2.4g routers(dual band could be better option but twice in price)
- Cat 6 cable (need site survey)
- Power cable
- Channels for Cable
- 4 4u Wall mount network rack (if required)

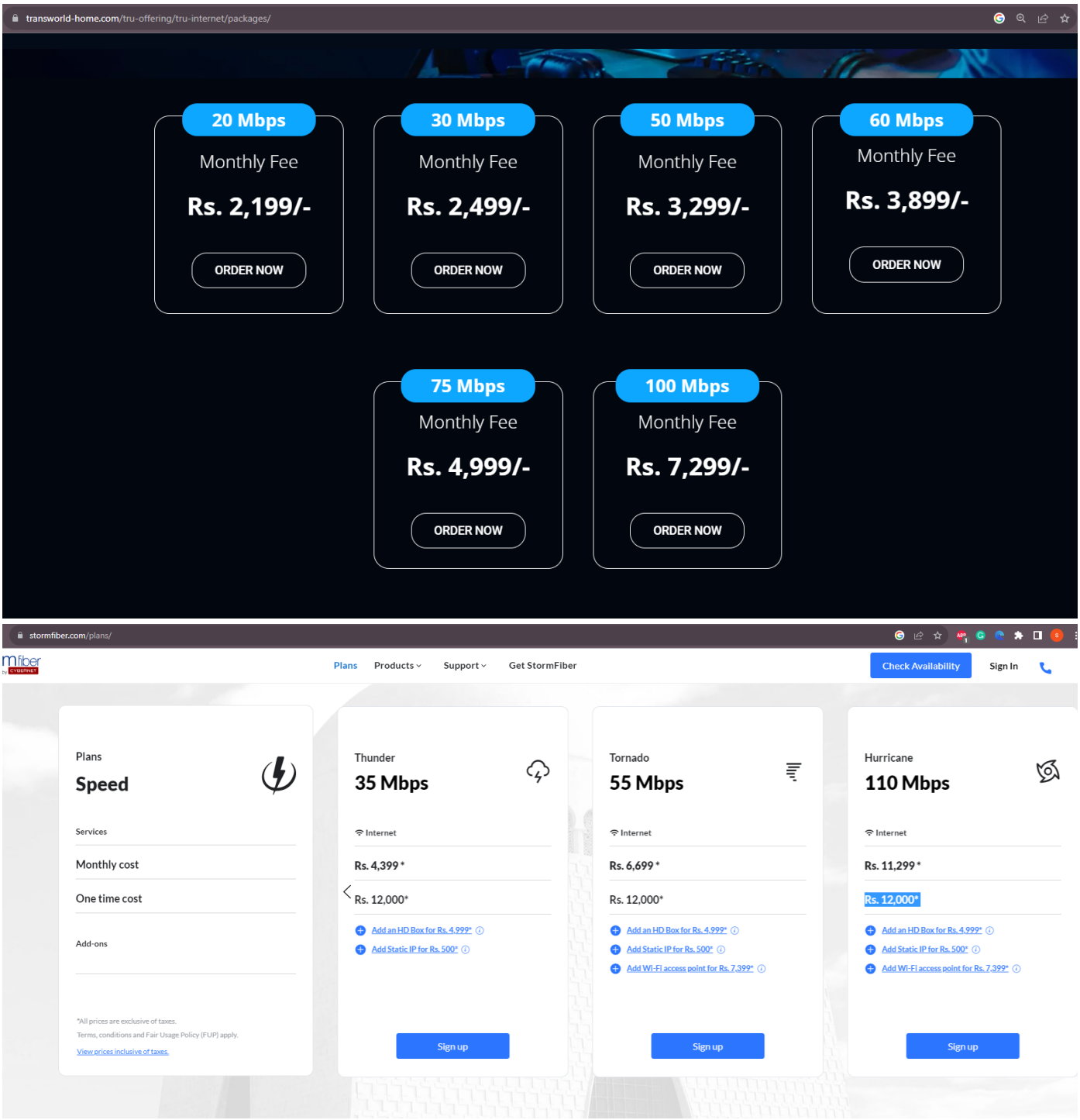
Cost Comparision

Item	Solution	Model/Specs	Qty	Estm Price	Total
Switch	Solution 01	Tp-Link SG1016D 16-Port Gigabit Rackmount Switch	12	18800	225600
Wifi Router	Solution 01	Exisiting routers will be utilized			
Cable	Solution 01	Required Survey			

Item	Solution	Model/Specs	Qty	Estm Price	Total
Channel	Solution 01	Required Survey			
Labour	Solution 01	Required Survey			
ISPs	Solution 01	Exisiting ISPs will be used	13		94000
Switch	Solution 02	48 ports cisco used switch	4	60000	240000
Wifi Router	Solution 02	TP link TL840 2.4G router	12	4500	54000
Cable	Solution 02	Required Survey			
Channel	Solution 02	Required Survey			
Labour	Solution 02	Required Survey			
ISPs	Solution 02	100mbps fiber internet connection	3	14500	43500
ISPs	Solution 02	35mbps fiber internet connection	1	5500	5500

Solution	SUM of Total
Solution 01	319600
Solution 02	343000
Grand Total	662600

Available Internet Plans in DHA Phase 2 Ext



Conclusion

The networking site survey indicates that converting the Wi-Fi-based network at ALMidrar to a wired network is a viable option. The transition offers significant benefits regarding network performance, reliability, and security. However, careful planning, resource allocation, and coordination with the IT team will be necessary to ensure a successful and smooth conversion process.